

Trade-Off Template

Day 1 handout (carried from Week 4). The standard shape for making a design choice explicit: Option A / Option B / Choice / Why / Accepted cost / Revisit trigger. Use it for schema choices this week — and for every architecture decision after.

Copy this block once per trade-off point in your design.

```
### Trade-off – <name of the choice>

**Tension:** <the two things in tension, e.g. Normalization vs denormalization>

**Option A:** <describe>
**Option B:** <describe>

**Choice:** Option <A/B>.

**Why:** <the reasoning, grounded in access patterns / SLOs / customer facts>

**Accepted cost:** <a specific thing you deliberately can't do
efficiently now – name the query or scenario>

**Revisit trigger:** <the concrete change that would make you
reopen this – a frequency change, a new customer, a
tightened SLO>
```

Three trade-offs every data model encounters

TRADE-OFF	TENSION
Normalization vs denormalization	Clean writes vs fast reads
Strong consistency vs replica reads	Always-fresh vs scalable read load
Schema flexibility vs schema integrity	Easy migration vs caught-by-database integrity

What separates a trade-off from a description

- A trade-off **names an Option B**. If there's no alternative, it's a description, not a trade-off.
- "Accepted cost" is a **named query or scenario** you deliberately can't run efficiently — not a vague word like "complexity."
- "Revisit trigger" is a **concrete, observable change** — usually a query-frequency shift or a new class of customer.

- At least one of your trade-offs should reference a **TCD §4 SLO**.